

Zhaofeng Luo

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Research interests: GPU-accelerated physics simulation; robust contact and deformable-body dynamics; numerical optimization; simulation for robotic manipulation.

Education

Carnegie Mellon University
Ph.D. student, Computer Science

Aug. 2025 – Present
Advisor: Prof. Minchen Li

Peking University
B.S., Computer Science

2021 – 2025
Beijing, China

Industry Experience

Genesis AI
Research Scientist Intern

2026
Physics simulation for robotics

Publications & Preprints

*Equal contribution.

- TapeSim: Efficient Simulation of Adhesive Tape Dispensing for Robotic Manipulation**
Zhaofeng Luo, Xinyu Lu, Jaehoon Choi, Zhehuan Chen, Trinity Chung, Xiaowen Qiu, Hugh Nicholas Perkins, Gianna Calderon, Alexis Duburcq, Sanghyun Son, Tsun-Hsuan Wang, Yi-Ling Qiao, Minchen Li.
Under review, IEEE ICRA 2027. [arXiv:2609.28766](https://arxiv.org/abs/2609.28766), 2026.
- AGIPC: Adaptive In-Solve Algebraic Coarsening for GPU IPC**
Xuan Wang, **Zhaofeng Luo**, Minchen Li, Taku Komura, Kemeng Huang.
ACM SIGGRAPH 2026 Conference Papers.
- Robust and Efficient Penetration-Free Elastodynamics without Barriers**
Juntian Zheng, **Zhaofeng Luo**, Minchen Li.
ACM Transactions on Graphics, 2026 (SIGGRAPH journal track).
- An Overlapping Schwarz Space-Time Refinement Framework for Material Point Method**
Zhaofeng Luo, Minchen Li, Yupeng Jiang.
[arXiv preprint arXiv:2605.09097](https://arxiv.org/abs/2605.09097), 2026.
- VR-Doh: Hands-on 3D Modeling in Virtual Reality**
Zhaofeng Luo*, Zhitong Cui*, Shijian Luo, Mengyu Chu, Minchen Li.
ACM Transactions on Graphics, 2025 (SIGGRAPH). Top 10 Technical Papers Fast Forward.
- Emulating Space Computing Networks with RHONE**
Liyang Wang, Qing Li, Yuhan Zhou, **Zhaofeng Luo**, Donghao Zhang, Shangguang Wang, Xuanzhe Liu, Chenren Xu.
USENIX Annual Technical Conference (ATC), 2025.
- A Self-Adaptive Retro-FSO Design for Air-to-Ground Communication**
Zhe Ou*, **Zhaofeng Luo***, Guanyu Shi, Chenren Xu.
ACM MobiCom 2023 Demo. Best Demo Candidate.

Selected Research

TapeSim: Adhesive Tape Simulation for Robotic Manipulation

CMU / Genesis AI

- First-author work on a simulator for dispensing, peeling, sticking, wrapping, and cutting adhesive tape, combining cohesive adhesion with adaptive reduction of the wound roll.
- For 32-turn rolls, mean physics steps were $4.5\text{--}8.4\times$ faster at comparable roll motion. Five real-world sticking replays had 23–29% lower mean image-plane core-landmark error than full-shell cohesive simulation.
- Demonstrated real-to-sim evaluation of frozen robot policies and teleoperated packaging with a robotic arm and dexterous hand.

GPU Solvers for Robust Contact and Elastodynamics

SIGGRAPH 2026

- **AGIPC** performs GPU algebraic coarsening within Newton solves, without changing mesh topology; up to $3\times$ speedup over StiffGIPC.
- **Barrier-free elastodynamics** uses an augmented-Lagrangian contact method to avoid barrier-induced ill-conditioning; up to $103\times$ speedup over GIPC on challenging contact-rich benchmarks.

Local Space-Time Refinement for MPM

arXiv preprint, 2026

- First-author work on overlapping Schwarz coupling of coarse and fine MPM subdomains through nonmatching-grid transmission and temporal subcycling; up to $9.15\times$ lower computational cost in the inclusion benchmark at comparable accuracy.

VR-Doh: Interactive Elastoplastic Modeling in Virtual Reality

SIGGRAPH 2025

- Co-first-author work on a real-time MPM-based system for natural hand-object interaction, sculpting, and deformation in VR, with localized simulation and particle-level collision handling.
- Combined physics-based interaction with 3D Gaussian Splatting to support appearance-preserving editing and simulation-ready asset creation.

Open Source & Technical Writing

Libuipc – Contributor to a C++20 GPU framework for unified incremental potential contact across rigid bodies, deformable solids, cloth, and threads.

Physics-based Simulation – Coauthor of the open online book with Minchen Li and Chenfanfu Jiang; developed the GPU solid-simulation tutorial and accompanying course material using CUDA/MUDA.

Teaching

Teaching Assistant, Carnegie Mellon University

Fall 2026

15-369/769: Numerical Methods: Foundations, ML, and Visual Computing

Teaching Assistant, Peking University

Sep. 2023 – Jan. 2024

Introduction to Computer Systems

Selected Honors

National Scholarship	2024
First Prize, Peking University Scholarship	2022, 2023
Gold Medal, Chinese Physics Olympiad	2020

Technical Skills

Programming: C/C++, CUDA/MUDA, Python, Rust, C#; Unity.

Simulation: Incremental potential contact (IPC), MPM, elastodynamics, contact and friction, GPU parallelization, sparse linear solvers, numerical optimization.

Robotics & interaction: Deformable-object manipulation simulation, teleoperation, physics-based VR interaction, 3D Gaussian Splatting.